

Unit 2, Lesson 3: Cube Values

Benchmark

MA.B.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.

Learning Targets

- ☐ I can use multiple representations to explore cube values.
- ☐ I can determine the positive and negative solutions for $x^3 = q$, where q is an integer.
- ☐ I can find the values of cube roots.

2.3.1 Warm-Up: Magic Square

1. Place the digits 1 to 9 in each square so the sum of the digits in each row, column, and diagonal is 15. Each digit can only be used once.

2.3.2 Exploration: Volume of Cubes

The volume of a cube can be found by multiplying the length by the width by the height. Volume is measured in cubic units.

Work with your partner to complete the following.

1. Complete the statements.

The length, width, and height of a cube have

- ☐ the same
- ☐ different

measure(s). Therefore, the volume of a cube can be found

by taking one side length to the

- ☐ first
- ☐ second
- ☐ third

power, which is

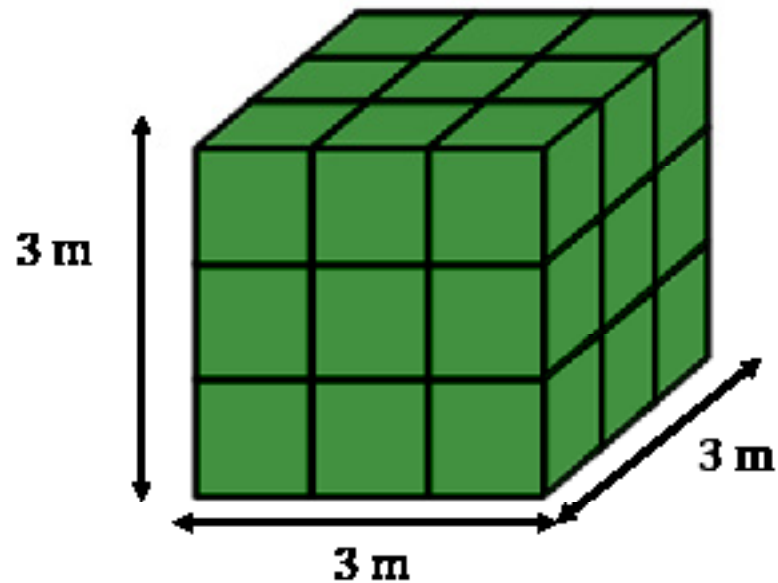
also known as

- ☐ squaring
- ☐ cubing

the side length.

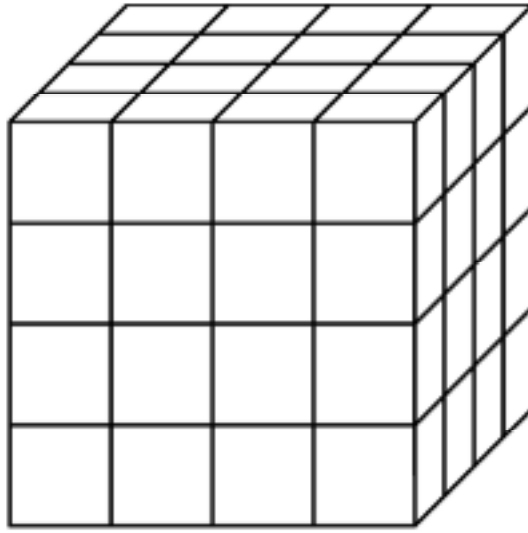
2. A cube is shown with dimensions given in meters (m).

Complete each statement.



- a. The edges of the cube each measure _____ m.
- b. The volume of the cube is _____ m^3 because _____³ = _____.

- 3. A cube is shown, where each gridline measures 1 unit.**



- a. Label the measures of the length, width, and height of the cube.**
- b. What is the volume of the cube?**
- 4. The numbers 27 and 64 are cube numbers, or perfect cubes.**
- a. Explain what you think a perfect cube is.**
- b. Work with your partner to find at least two other numbers that are perfect cubes.**

2.3.3 Bring It Together: Perfect Cubes

1. Discuss with your partner the similarities and differences between square numbers and cube numbers. Then, summarize your discussion in the table.

Similarities	Differences

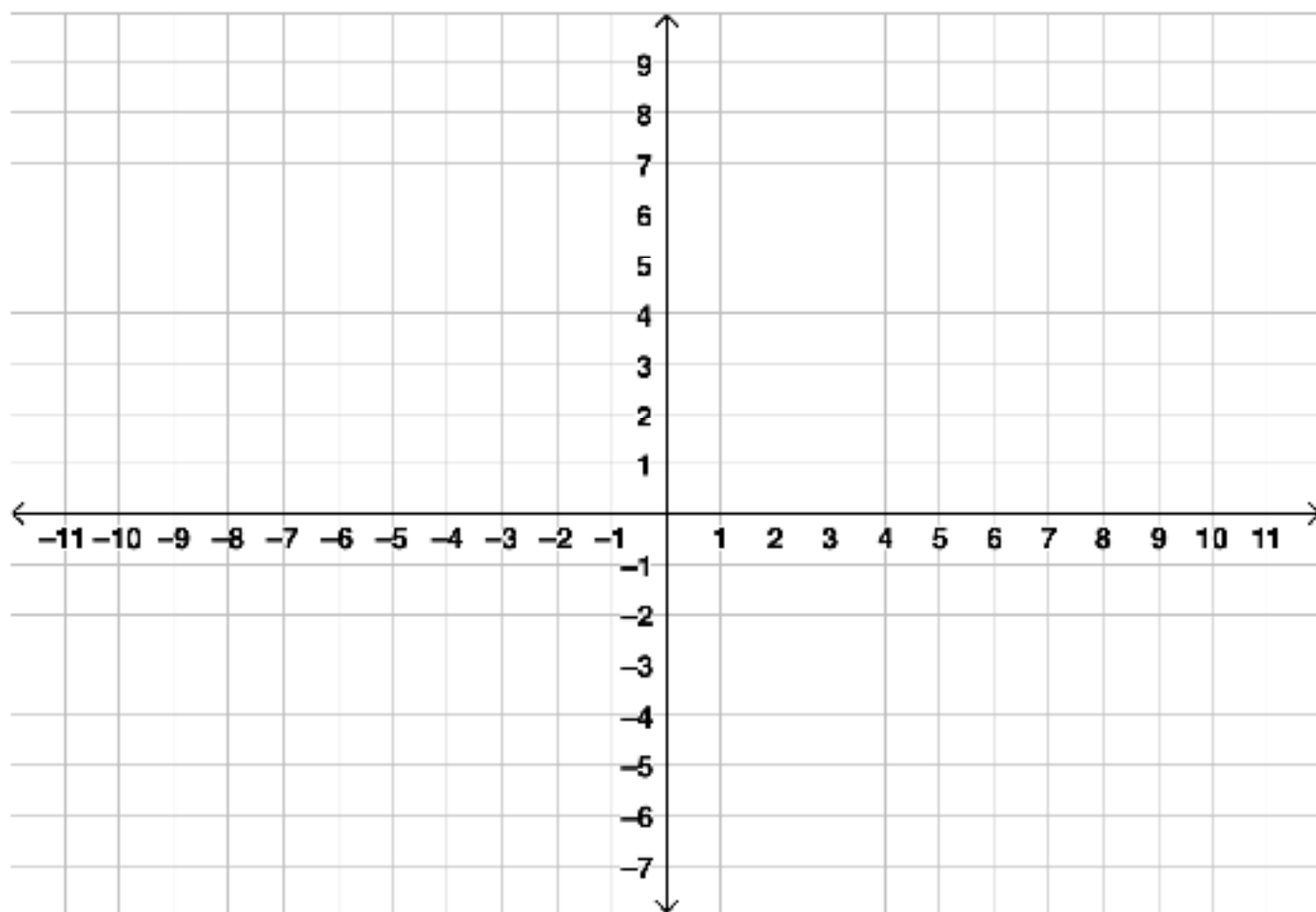
2.3.4 Exploration: Cube Values on the Coordinate Plane

1. Use the equation $y = x^3$ to complete the following with your partner.
 - a. Complete the table of values for the equation $y = x^3$ by substituting the given values of x and solving for the corresponding y -value.

x	Substitute into $y = x^3$	Coordinates (x, y)
-2	$y = (-2)^3$ $y = \underline{\hspace{2cm}}$	$(-2, \underline{\hspace{2cm}})$
-1		
0		
1		
2		

- b. The y -values are cube numbers. What do you notice about the y -values?

c. Plot the coordinates on the grid shown.



Notice the graph of $y = x^3$ includes points above and below the x -axis, unlike the graph of $y = x^2$ explored in a previous lesson.

2.3.5 Guided Instruction: Solving Equations in the Form $x^3 = q$

1. Find the solution to each equation.

a. $x^3 = 8$

b. $-8 = x^3$

2. Discuss with your partner the similarities and differences between the equations and their solutions.

3. Complete the statements.

For equations in the form $x^3 = q$, where q is an integer,

there

- ☐ is only one solution.
- ☐ are two solutions.
- ☐ are three solutions.

The solution(s) can

- ☐ only be positive.
- ☐ only be negative.
- ☐ be positive or negative.

The sign of the solution(s)

depends on the sign of q .

Similar to the relationship between squares and square roots, there is an inverse relationship between cubes and cube roots.

The symbol $\sqrt[3]{}$ is used to indicate a **cube root**, where **3** is in the index of the radical.

Cube roots can be used to solve equations in the form $x^3 = q$, where q is an integer.

4. Complete the table for each equation.

Equation	Related Cube Root	Solution
$x^3 = 27$	$\sqrt[3]{27} = x$	$x = \underline{\hspace{1cm}}$
$p^3 = -1$	$\sqrt[3]{} = p$	$\underline{\hspace{1cm}} = p$
$\underline{\hspace{1cm}} = s^3$	$\sqrt[3]{64} = s$	$s = \underline{\hspace{1cm}}$
$w^3 = \underline{\hspace{1cm}}$	$\sqrt[3]{} = w$	$w = 6$
$\underline{\hspace{1cm}} = c^3$	$\sqrt[3]{-27} = c$	$c = \underline{\hspace{1cm}}$
$-125 = b^3$	$b = \sqrt[3]{}$	$\underline{\hspace{1cm}} = b$

2.3.6 Wrap-Up: Ticket Out the Door

1. Find the value of each of the following.

a. $\sqrt[3]{125} =$

b. $\sqrt[3]{-8} =$

2. Solve each equation.

a. $w^3 = -64$

b. $1 = k^3$

